

Are the Skilled Trades Safe From AI?

The complete profile — the full scores by track, the six factors behind them, the honest downsides, and what to ask before you commit.

2.5

AI EXPOSURE · ELECTRICIAN — SERVICE & MAINTENANCE
where 10 is most at risk

Read this one first; send the student version to them.



Before you read this — a note for parents

You're likely reading this before your child is. That's normal, and it's worth pausing on how you use it.

Don't lead with the scores. If you open with a risk number, they either get frightened and shut down, or defensive about a decision they've already made.

What this is. A facts document, not a recommendation.

A practical note: there's a separate short version written directly to them. Send them that one.

And two things specific to the trades, because this profile carries more baggage than any other in the index.

First, for many parents this is the profile they least expected to be reading. If your child has raised the trades and your instinct was disappointment, that reaction is worth examining before the conversation rather than during it. The scores here are better than most degrees, the debt position is better than all of them, and the demand picture is stronger than anything else we measure. The disappointment is usually about status rather than outcomes.

Second, and equally honest: this is not a consolation prize and it is also not for everyone. The physical demands over thirty years are real, the income ceiling without business ownership is real, and a teenager who dislikes being outdoors in February will dislike it for a very long time. Good news should open the conversation, not close it.

The short answer

For the right person, this is the strongest bet on the board — and almost nobody considers it seriously.

A service electrician scores 2.5 out of 10 — among the lowest scores in this entire index. No tuition debt. An on-ramp that is *widening* while most degree routes narrow. And the demand picture is the strongest we measure anywhere.

The protection was never "working with your hands." It was working with your hands somewhere unpredictable. That distinction is the whole profile, and it explains why the same

skills on a production line score twice as high.

The scores

TRADE AND SETTING	2023	2025	FALL 2026	3-YR MOVEMENT	BAND
Electrician — service & maintenance	2.1	2.3	2.5	+0.4	Very low
Plumbing / HVAC — service	2.2	2.4	2.6	+0.4	Very low
Welding & fabrication — site work	2.8	3.0	3.2	+0.4	Low
Construction site trades	2.8	3.1	3.4	+0.6	Low
Production line / prefab / modular	4.0	4.5	5.0	+1.0	Moderate

Scores run 1–10, where 10 is most at risk. For context: the median profession in this edition sits around 5.5, bedside nursing scores 2.8, licensed engineering 4.0, a business analyst 7.6, and entry-level software development 8.1.

Read the table this way: the service electrician's +0.4 over three years is the slowest movement in the entire index. Entry-level software moved +1.8 over the same period. And note the last row — the same trade skills in a factory setting move at more than twice the rate of the same skills in the field.

Why the trades score so low — the six factors

Here's how a service electrician answers them.

HOW MUCH OF THIS JOB CAN AI ALREADY DO? (35% OF THE SCORE)

The paperwork, and essentially nothing else.

Quoting and estimating. Scheduling and dispatch. Invoicing and job admin. Code and regulation lookup. Parts ordering and inventory.

For a self-employed tradesperson, almost all of that is a benefit rather than a threat. The office burden of running a trade business is the part being automated.

What resists: diagnosing a fault in a sixty-year-old building, running new service through a wall with no drawings, working in an attic or a crawlspace or a trench, and judging whether conditions are safe to continue.

HOW HARD WILL IT BE TO LAND THAT FIRST JOB? (15%)

Among the best on-ramps we measure anywhere, and it is getting better rather than worse.

Apprenticeships are paid rather than charged for, legally embedded in licensing, and chronically undersubscribed. The route in is widening at exactly the moment most degree routes are narrowing.

BLS projects **649,300 annual job openings** in construction and extraction occupations over 2024–2034. Electricians specifically show **9% growth** — classified as "much faster than average" — with roughly **81,000 openings each year** against total employment of about 818,700.

DOES IT HAVE TO BE DONE IN PERSON, WITH YOUR HANDS? (15%)

Completely, and in the hardest possible setting for automation.

This is the factor that does most of the work, and section 5 explains why "physical" is not one thing.

DOES SOMEONE NEED A HUMAN THEY CAN TRUST AND HOLD RESPONSIBLE? (15%)

Yes, and more than people assume.

Someone is being let into a home. Someone is signing off that the work is safe. When an installation fails and causes a fire or a flood, a licensed human is answerable.

DOES THE LAW REQUIRE A LICENSED HUMAN? (10%)

For electrical and gas work, comprehensively.

Licensing, inspection and certification requirements are written into building codes and enforced by inspectors. Nobody signs off their own unlicensed electrical work.

HOW OFTEN DOES THE JOB HIT GENUINELY NEW, HIGH-STAKES SITUATIONS? (10%)

Every single job.

A different building, unlabelled wiring, work someone else did badly in 1974, and a customer describing the problem inaccurately. The environment is different every time, which is precisely the condition robotics handles worst.

"Manual work is safe" is the wrong idea

This deserves its own section, because the popular version of this argument is wrong in a way that matters — and the trades are where it breaks most visibly.

The usual framing is binary: desk jobs are exposed, manual jobs are safe.

What actually matters is not whether the work is physical. It's whether the environment is predictable.

Robotics performs well in controlled settings — known dimensions, repeated tasks, fixed tolerances, no surprises. That is a warehouse, an assembly line, a modular construction facility. It performs badly in uncontrolled ones, and that gap is not closing quickly.

A service electrician spends the day in unpredictable environments by definition. That rates **9.5** on our physical factor.

The same trade skills applied to a production line rate **8.0** — still physical, but in exactly the setting robotics handles best. That single difference, plus weaker licensing and lower trust requirements, is what separates a 2.5 from a 5.0.

So the useful question for any hands-on career is not "is this manual work?" but "would two days of this job ever look the same?" If the answer is no, the protection is real. If the answer is yes, physical work offers much less shelter than people assume.

This is also the thing to watch over a career. Modular and offsite construction is growing, and it works by moving site work into factories. It is the one development that could genuinely move these scores — not because robots got better at crawlspaces, but because the industry moved the work somewhere robots already function.

What the trend shows

Service electrician: 2.1 → 2.3 → 2.5. Up 0.4 in three years — tied for the slowest movement in the entire index, and all of it administrative.

Production line / prefab: 4.0 → 4.5 → 5.0. Up 1.0, more than twice the pace.

The gap widened from 1.9 points to 2.5. The pattern is now familiar: the unpredictable end holds while the structured end drifts. What is unusual here is how little the protected end has moved at all.

How durable is the protection?

PROTECTION	STRENGTH TODAY	DURABILITY	WHERE THE PRESSURE IS COMING FROM
Physical / embodied	Strongest in the index	The one to watch	Not from robots reaching crawlspaces, but from prefabrication moving work into factories.
Regulatory / licensing	Strong for electrical and gas	Most durable	Building codes and inspection regimes move at the speed of legislation.
Judgment under novelty	Very strong	Durable	Every building is different. No two service calls are alike.
Human trust / accountability	Strong	Durable	Someone is in a home, and someone signs off on safety.

The honest read for a seventeen-year-old. They'll work for roughly forty-five years — longer than most graduates, because they start earlier. Over that span, the licensing and novelty protections should hold comfortably. The physical protection is the strongest today and the one with the most credible long-term threat — and the threat is industrial reorganisation rather than robotics.

How the role will actually change

WAS THE JOB	IS BECOMING THE JOB
Working out the quote by hand	Software quotes it; you check it's right
Ringling round for parts	Ordering is automated
Paperwork in the evening	Largely gone
Diagnosing from experience	Diagnosing from experience, plus better instruments
Learning codes by memory	Looking them up instantly on site
The actual work	Unchanged

The last row is the point. In most professions the automatable layer and the skilled layer are tangled together. In the trades they are cleanly separable, and what's automating is the part nobody enjoys.

Human strengths that will matter most

1. **Diagnosis in unfamiliar conditions** — working out what's wrong in a building nobody

documented. The core skill and the least automatable thing here.

2. **Physical judgment** — knowing whether something is safe to proceed with, by feel and by

eye.

3. **Being trusted in someone's home** — a genuine commercial asset, and the foundation of a

business.

4. **Customer handling** — explaining a problem plainly to someone worried about the cost. 5. **Business sense** — the thing that separates a \$60,000 tradesperson from a \$200,000 one.

Other things to consider

What's genuinely good about it

The financial position is better than most families realise, and the comparison usually never gets made.

- **No tuition debt.** Apprentices are paid from day one — typically starting at \$15–20 an hour and stepping up annually.
- **Four to five years to journeyman status,** earning throughout.
- **BLS median wages:** electricians **\$62,350**, plumbers **\$62,970**, with top earners in both exceeding **\$106,000**.

- **Union total compensation runs roughly 60% higher** than non-union once health, pension and benefits are counted, per BLS.
- **Construction wages grew 4.2% year-over-year** into mid-2025, outpacing the national average across all occupations.

The comparison worth running: a graduate finishes at 21 or 22 with debt and enters a market where 41.5% of recent graduates are underemployed. An apprentice finishes at 22 or 23 with no debt, four years of earnings behind them, a licence, and — currently — employers competing for them.

And what tradespeople report as rewarding is consistent and distinctive: the problem is solved and visible by the end of the day, the work is genuinely varied, autonomy is high, and running your own business is a normal outcome rather than an exotic one.

What's genuinely hard about it

The physical toll over decades is the real cost, and it is the thing most likely to end a career early. Knees, backs and shoulders. The work is done in heat, cold, confined spaces and awkward positions.

To be fair to the data: the nonfatal injury rate for specialty trade contractors is **2.3 per 100 full-time workers** and has been declining, driven by sustained focus on falls, struck-by, electrocution and caught-in hazards. It is safer than it was. It is still harder on a body than any desk job, and that compounds over thirty years.

The income ceiling without ownership is real. Median pay is solid and the top end is excellent, but the path to the top end usually runs through running a business — which is a different skill from the trade itself, and not everyone wants it.

It is cyclical. Construction rises and falls with credit conditions and public spending in a way that nursing and teaching do not.

And the status question is real, even if it shouldn't be. Some people find the condescension wearing, particularly from people who assume it was a fallback.

Who this work suits — and who it doesn't

The trades tend to suit people who:

- **Want the problem solved by the end of the day.** The single most-cited satisfaction.
- **Would rather be moving than sitting**
- **Like diagnosis** — the puzzle of working out what's actually wrong

- **Are practical with people** — most trades involve customers, not just materials
- **Are interested in eventually running something**

It's harder for people who:

- Genuinely **dislike physical discomfort**. Being outdoors in February is not a phase.
- Want **predictable indoor conditions**
- Need **structured career progression** with titles and ladders
- Are **conflict-averse about money** — chasing invoices is part of self-employment

What else is moving this market

The shortage is severe and well documented. The construction industry needs to attract hundreds of thousands of net new workers in 2026 alone, against an ageing existing workforce.

And here is the detail that reframes this entire profile. Microsoft's president Brad Smith has publicly identified the electrician shortage as the single biggest constraint on data centre expansion — not chip supply, not land, not permitting. Industry estimates put the requirement at 300,000-plus additional electricians just to meet AI-related data centre demand. Electricians working on data centre projects report earnings well into six figures.

Capital has noticed: BlackRock launched a \$100 million initiative in March 2026 aimed at training 50,000 skilled trade workers, and the US Department of Labor announced \$145 million in apprenticeship grants targeting shipbuilding, defence, semiconductors and energy.

So the AI buildout is simultaneously the thing threatening white-collar entry work and the thing driving demand for electricians. That is not irony — it is the most concrete illustration in this index of where the protection actually lies.

One honest caution. A meaningful share of current construction demand is project-driven — data centres, infrastructure programmes — and those end when the builds do. The demographic case, an ageing workforce retiring faster than replacements arrive, holds regardless of the cycle. A student should weight the demographics more heavily than the boom.

The AI-native advantage — what to actually do about it

The trades are unusual here: AI helps a tradesperson mainly by removing the parts of the job that aren't the trade.

1. Run the business side properly. Quoting, scheduling, invoicing and customer follow-up are now largely automatable. The tradespeople who use this well aren't better at the trade — they're better at running a trade business, which is where the income ceiling actually sits.

2. Use diagnostic tools without deferring to them. Thermal imaging, circuit analysis and fault-finding tools are improving. They inform judgment; they don't replace it.

3. Look things up rather than memorise them. Codes change constantly. Instant lookup is a genuine efficiency gain.

4. Specialise deliberately. Data centres, EV infrastructure, solar and battery, heat pumps. These are where the premium is and where demand is growing fastest.

5. Get to ownership. Everything in this profile points at it. The trade gives you a protected skill; the business is what converts it into a ceiling.

Concrete preparation: get any hands-on experience before committing — a weekend with someone working is worth more than any brochure. Compare apprenticeship providers on completion rates and on what happens after qualification. And choose the trade deliberately: electrical leads on pay, growth and protection, and it is the one sitting directly in front of the AI-driven demand wave.

The routes in

ROUTE	NOTES
Registered apprenticeship	The main route. Paid from day one, 4–5 years to journeyman, nationally recognised credential.
Union apprenticeship	Structured, well-paid, with substantially higher total compensation. Competitive entry.
Pre-apprenticeship programme	For someone with no experience. Short, and a route into the above.
Trade school then apprenticeship	Faster classroom start, but costs money the apprenticeship route doesn't.
Military technical training	Excellent trade grounding with civilian licensure pathways.

The thing to compare: completion rates and post-qualification placement, not marketing.

Where this can lead later

The ceiling is higher than most parents assume, and it is reached differently.

- **Journeyman → master → contractor** with your own licence
- **Business ownership** — the main route to the top end
- **Site supervision and construction management** — trade-to-supervision remains a fully respected path, and construction management scores 3.3
- **Specialist and high-demand work** — data centres, industrial, marine, controls
- **Inspection, code enforcement and safety**
- **Teaching the trade** — instructor roles at colleges and apprenticeship providers
- **Estimating and project management** — note this moves toward the desk, and the score climbs accordingly

The pattern holds here as everywhere: moving away from the physical work raises exposure. Estimating scores 6.3. It's a good career and a different one.

What to look for in an apprenticeship

1. **Completion rate.** The number that matters most and the one least advertised.
 2. **Breadth of work.** Domestic only, or commercial and industrial too? Breadth determines the ceiling.
 3. **What happens at the end.** Is there a job, and at what rate?
 4. **Licensing pathway.** Does the programme lead cleanly to the licence, or leave gaps?
 5. **Quality of supervision.** In an apprenticeship, the person teaching you is the education.
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Questions to ask an apprenticeship provider or employer

- What proportion of apprentices complete, and what happens to those who don't?

- What's the pay progression year by year, and what do journeymen here earn?
- What range of work will I see — domestic, commercial, industrial?
- Do you have a job at the end, and what proportion of apprentices stay on?
- What licence does this lead to, and is anything else required?
- Who supervises apprentices day to day, and how many do they carry at once?

Red flags: vagueness about completion rates; narrow work only; no clear licensing pathway; apprentices used as unsupervised labour.

Go deeper — further reading

- **Electricians — Occupational Outlook Handbook** — US Bureau of Labor Statistics. Primary source and free: 9% projected growth through 2034, roughly 81,000 annual openings, wage distributions and entry requirements. Look up plumbing and HVAC while you're there.
- **Electrician career guide 2026** — the data centre story in detail, including Microsoft's identification of the electrician shortage as the primary constraint on expansion, and the industry estimate of 300,000-plus additional electricians needed.
- **Skilled trades statistics for 2026** — wages, injury rates, union compensation differentials and the BLS opening projections collected in one place.

The counterargument — read this one:

The strongest case against our score isn't that robots will reach crawlspaces. It's **prefabrication**.

Modular and offsite construction is growing steadily, and it works precisely by moving work from unpredictable sites into controlled factories — the setting where our own framework says automation performs best. On this reading, the trades' protection erodes not because the work becomes automatable, but because the industry relocates it somewhere it already is.

Our own table concedes the mechanism: production-line and prefab work scores 5.0 against 2.5 for field service. If the share of construction done offsite rises substantially, the average tradesperson's exposure rises with it — and our headline figure would be describing a shrinking share of the trade.

Where we land: we hold the score because service, maintenance and retrofit work — which is a large and growing share of trade employment, driven by an ageing building stock — cannot be prefabricated. You cannot factory-build a repair to a 1963 house. But we'd treat rising offsite construction share as the single most important thing to watch in this profile, and we'll report it if it moves.

Bottom line

For the right person this is the strongest position in the index, and it is chronically under-recommended to exactly the students who would thrive in it.

The scores are among the lowest we measure. There is no tuition debt. The on-ramp is widening while most degree routes narrow. Demand is severe, well-documented, and — in electrical work — being driven by the AI buildout itself.

The protection is specific and worth understanding rather than assuming. It is not that the work is manual. It is that **the environment is different every time**, which is the condition automation handles worst. Move the same skills into a factory and the score doubles.

The honest costs are real and have nothing to do with technology. Physical wear over decades is the thing most likely to end a career early. The income ceiling without business ownership is genuine. And construction is cyclical in a way that healthcare and education are not.

So the useful advice is: choose the trade deliberately rather than taking whichever apprenticeship appears first, aim at service and retrofit work rather than repetitive new-build, compare providers on completion rates, and plan for ownership from the start rather than discovering at 35 that the ceiling is lower than expected.

And for a parent whose first reaction to this was disappointment — it's worth sitting with where that came from. On the evidence in front of us, this is a better financial and occupational position than most degrees currently offer.

Discussion questions

Part 1 — About the work itself

1. What made the trades appealing? Push past "I'm practical" to something specific. 2. Do they like *diagnosis* — working out what's wrong — or *building*? Different trades reward each differently. 3. Have they done anything hands-on for a sustained period? What did they think? 4. How do they actually feel about being cold, wet or in a confined space? Not in theory — in practice.

Part 2 — Reacting to the findings

5. A service electrician scores 2.5 and an entry-level software developer 8.1. Was that surprising? 6. Production line work scores twice as high as field service. Does that change which trades or settings appeal? 7. The profile argues the protection is unpredictability rather than manual work. Does that land?

Part 2b — The harder conversation

8. **The physical toll over thirty years.** Have you talked about what that actually means at 50? 9. The income ceiling usually runs through owning a business. Does that appeal, or is it a chore? 10. Construction is cyclical. What's the plan in a downturn?

Part 2c — The other side

11. Of what tradespeople find rewarding — the problem solved by day's end, variety, autonomy, ownership — which two matter most? 12. Looking at "who this work suits": which items sound like them? Answer separately, then compare.

Part 2d — The money question

13. Have you actually compared the numbers? Apprentice earning from 18 with no debt, against a graduate starting at 22 with debt in a market where 41.5% of graduates are underemployed. 14. Do they know union apprenticeships carry roughly 60% higher total compensation? Have they looked at what's available locally?

Part 3 — Testing it against reality

15. **The most valuable single action:** spend a day with someone working. Not a college open day — an actual working day, including the unglamorous parts. 16. Find someone ten years in and

ask what their body feels like, and whether they'd do it again.

Part 4 — About the programme

17. Ask each provider the completion-rate question from section 13. The evasiveness tells you plenty. 18. Is there a job at the end, and what proportion stay on?

Part 5 — Widening the frame

19. Which trade, and why? Electrical leads on pay, growth and protection — has that been considered specifically? 20. Have they looked at construction management? Trade-to-supervision is a normal route, and it scores 3.3. 21. If the trades weren't available, what would they choose? The answer usually reveals what they're actually drawn to.

Part 6 — For the parent, alone

22. **What was my honest first reaction when they raised this?** Where did that come from? 23. Am I comparing this to a degree on outcomes, or on status? 24. Have I actually looked at what apprentices earn, and what journeymen earn, or am I assuming? 25. What's the one thing I most want them to understand — in a sentence, without a number in it?

This is analysis and informed judgment about a fast-moving situation, not a guarantee or a prediction. For the trades specifically, we've flagged that the most credible long-term threat is prefabrication moving work into controlled settings, rather than robotics reaching uncontrolled ones.

Scores for 2023 and 2025 are retrospectively calculated using the current methodology. Scores measure exposure to what AI can already do — not how much any particular employer has deployed.

Appendix — technical scoring

The factor scores behind the headline number, shown for the **Electrician — service & maintenance** track. Each factor is rated 0–10 and carries a fixed weight; exposure factors push the score up, protection factors pull it down. Full method at pivotum.ai/methodology.

FACTOR	WEIGHT	RATING	EFFECT
How much of this job can AI already do?	35%	5.0	Exposure ↑
How hard will it be to land that first job?	15%	2.0	Protection ↓
Does it have to be done in person, with your hands?	15%	9.5	Protection ↓
Does someone need a human they can trust and hold responsible?	15%	8.5	Protection ↓
Does the law require a licensed human?	10%	9.0	Protection ↓
How often does the job hit genuinely new, high-stakes situations?	10%	9.0	Protection ↓

Scores run 0–10, where 10 is most exposed to what AI can already do. Re-scored every six months.